TED (15) -4014 Reg.no

Revision -2015 Signature:

FOURTH SEMESTOR DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY

**THEORY OF STRUCTURES -II**

(Maximum mark:100) (Time: 3hours)

PART-A (Maximum mark: 10)

I. Answer the following questions in one or two sentences. Each question carries 2 marks

1). Define ‘short column’

2). What are the different end conditions of a column?

3).What are the conditions of stability of a dam?

4).Write the equations for determining the maximum slope and deflection of a cantilever beam of span l with point load’ W ‘ acting at free end.

5).What is distribution factor?

PART-B (maximum mark: 30)

11. Answer any five questions from the following. Each question carries 6 marks.

1). State the difference between’ perfect frame’ and’ imperfect frame’

2). Write the limitations of Euler’s formula for critical load

3).A solid rectangular column 300mm wide and 200 mm thick carrying a vertical load of 25KN at an eccentricity of 50mm in plane bisecting thickness. Determine maximum and minimum intensities of stress in the section

4).Write three advantages and disadvantages of a fixed beam over simply supported beam.

5).A rectangular R.C simply supported beam of span 2 meter and cross section 200x250mm is carrying a uniformly distributed load of 10KN/m throughout the span. Find the maximum slope and deflection at the free end. Take E =$2x10^5$ $n/mm^2$

6).A cantilever beam AB of length l carries a point load of W at its free end. Using Mohr’s theorem find the slope and deflection at the free end.

7). Define and derive the equations for ‘stiffness factor ‘ ofa beam having simply supported ends andone end fixed and other end simply supported.

PART-C (maximum mark :60)

UNIT-1

111. a) A Built up column has a length of 5 m with one end fixed and other end hinged. Calculate the safe axial load by Rankine's formula and a factor of safety 3.

Least moment of inertia= $440.40 \times 10^4 mm^4$

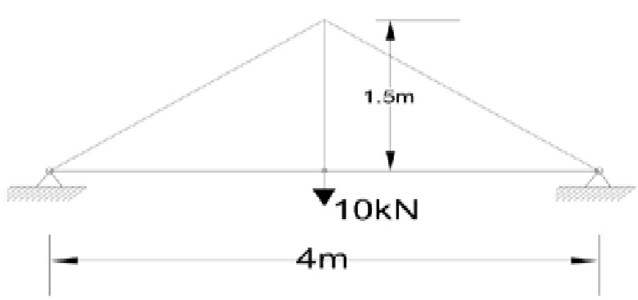
Cross sectional area = $5047 mm^2$

Rankine's constant $\propto$ =1/7500 and $f_c$= 320n /$mm^2$ (7)

b) Derive Rankine's formula for columns (8)

OR

1V. Using method of joint, determine the forces in the members of a truss shown below (15)




Unit-11

V. a) Derive the maximum possible value of eccentricity in case of a circular section, so that only compressive stress is developing at the base of the column, under eccentric loading. (7)

b) Amasonry trapezoidal dam 5m height,1m wide at its top and 3 m wide at the bottom retains water on its vertical face. Determine the lateral thrust of water, weight of dam, resultant force and the position of all the above forces, when the reservoir is full. Density of masonry is $19.62 KN/m^3$. (8)

OR

VI a)A solid circular column 200mm in diameter carries an eccentric load which produces a uniformly varying stress from zero at one edge to 125n/$mm^2$at the opposite edge. Find the eccentric load and the eccentricity (7)

b).Calculate the fixing moment values of a fixed beam with a span of 3.60m and carrying a uniformly distributed load of 14KN/m, for the entire span. Draw the SF and BM diagrams of the beam under this loading (8)

UNIT-111

V11. a)Derive the expression for the slope and deflection of a simply supported beam of span l , carrying a point load of W at the centre, by using Mohr's theorems (7)

b) Calculate the maximum deflection at the centre of a simply supported beam of span 4.2m, when its maximum slope is $3^0$ (8)

OR

V111. a) Derive the basic deferential equation of the elastic curve (7)

b) A cantilever beam of uniform flexural rigidity and 5m span carries a point load of 50KN at its midspan, along with a uniformly distributed load of10KN/m from the mid span to the free end

Take EI=40000KN/$m^2$, determine the slope and deflection at the centre of the span using Moment area method (8)

UNIT-1V

1X A continuous beam ABC is simply supported at A, B and C having a span AB= 6m and BC= 4m

The span AB carries a point load of 3KN at 2m away from the support A, the span BC carries a Uniformly distributed load of 1KN/m. Find the reaction and bending moment at supports A,B and C. Also draw the SF and BM diagrams (15)

OR

X A continuous beam ABCD is fixed at A, simply supported at B,C and free at D. AB=BC=5m and CD=2m. A uniformly distributed load of 15KN/m is acting in span AB and BC and appoint load Of 20KN at D. Determine the support moment by moment distribution method and draw the bending moment diagram. (15)